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[18MSCH2H01]

SECOND SEMESTER UNIVERSITY EXAMINATION, MAY - 2019

INORGANIC CHEMISTRY - II

Time: 03 Hours **Maximum Marks:70**

Instructions: I. **Question 1 is compulsory, carries 10 marks.**

II. **Questions 2 to 7 carry 15 marks each, answer any four.**

1. Answer any FIVE. **5×2=10**

a. Define ambidentate ligand with examples.

b. Deduce the relationship between Δ_o and Δ_t.

c. Define Fischer and Schrock carbenes.

d. What are f–f and d–d transitions?

e. Define Curie temperature.

f. State 18–electron rule with an example.

2. a. Explain geometrical and coordination isomerisms with suitable examples.

b. Write a note on stability constants of coordination compounds with illustrations.

c. Name the following compounds as per the IUPAC nomenclature;

i. $Cs_3[FeCl_6]$,

ii. $[Ru(NH_3)_4Cl(H_2O)](ClO_4)_2$, and

iii. $[Pd(PPh_3)_4]$ **(5+4+6)**

3. a. Explain the bonding involved in $[Fe(Ox)_3]^{3-}$ and $[Ir(NH_3)_6]^{3+}$ using VBT.

b. Write the crystal field splitting pattern for $[Co(NH_3)_6]^{3+}$ and $[Ni(CO)_4]$.

c. Write brief notes on:

i. spectrochemical series of ligands

ii. MOT of $[CoF_6]^{3-}$. **(5+5+5)**

4. a. Illustrate the different modes of coordination of dihydrogen and dinitrogen in transition metal complexes with suitable examples.

b. Explain the bonding in acylic carbene, N-heterocyclic carbene and carbyne coordinated transition metal complexes.

c. Write a brief note on the use of circular dichroism in measuring optical activity of coordination compounds. **(5+5+5)**

5. a. Write a note on Orgel diagrams considering d^1, d^4, d^6 and d^9 cases as examples.

b. Calculate the ground state term symbol for Cr^{3+}, V^{3+}, Mn^{2+}, Co^{2+} and Zn^{2+}.

c. Determine the term symbol for d^4, d^6 cases and identify the ground state term. **(5+5+5)**

$2(2)+1 = 5D$

$L = 4+1-1-2 = 2$ D.

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6. a. What is spin cross over in coordination compounds? Explain with appropriate examples.

b. Calculate the spin only magnetic moment for $K[AuCl_4]$, $[Cu(NH_3)_4]SO_4$, and $Mg[Zn(CN)_4]$.

c. Explain the significance of orbital contribution in calculating effective magnetic moment of a complex molecule. **(5+6+4)**

7. a. Explain the classification of organometallic compounds.

b. Explain the following:

i. fluxional behaviour of organometallic compounds, and

ii. isolobal principle

c. i. Calculate the number of electrons with central metal ion in, $[Ni(PPh_3)_2(CO)_2Cl_2]$.

ii. Calculate the number of bonds between metal centers in $[Co_4(CO)_{12}]$.

(5+5+5)